

## ④ Correlation & experimentation

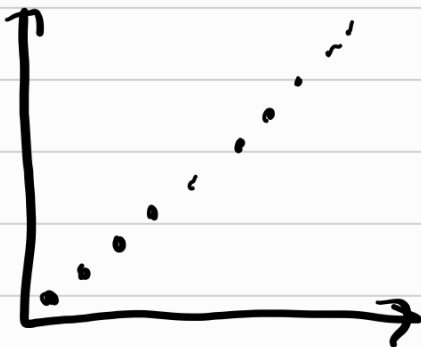
How do variables relate?

Correlation refers to the relationship between 2 variables

Correlations can be positive or negative

The correlation coefficient,  $r$ , ranges from -1 to 1

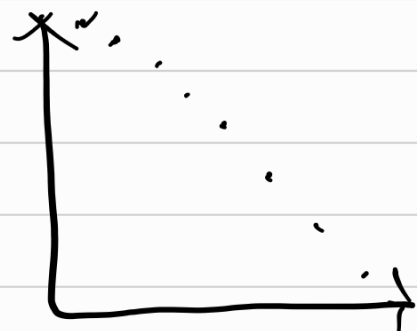
Scatterplots are graphic representations that show correlation between 2 variables



perfect positive  
correlation  
 $r = 1$



no  
relationship  
 $r = 0$



perfect negative  
correlation  
 $r = -1$

Directionality problem:

Correlation  $\neq$  Causation

Personal Example:

if someone were to measure my average sleep in a week and my avg caffeine intake they would find a negative correlation.

But not a causation. Maybe I drink more caffeine because I'm tired or caffeine is hurting my sleep

3rd variable problem:

there may be an unaccounted for variable

Are all correlations we see real?

An Illusory correlation is a perception of a relationship between 2 variables when no such relationship exists

Fortunately Regression to the mean means that extreme results eventually move close to average after many tries

# How do scientists isolate cause & effect?

Researchers use experiments to identify cause & effect.

In experiments, researchers manipulate or measure independent variables, holding other factors constant, to observe the effect on dependent variables.

Researchers create an experimental group & a control group

To minimize preexisting differences, researchers randomly assign people to each group

## Procedures & Placebo

The Placebo effect is a very powerful effect where people can experience real effects from a treatment simply by believing that it's effective

To avoid biases, like Placebo, many experiments are done single or double blind to reduce experimenter bias

